TITLE PAGE

- **Problem Statement ID -** SIH26045
- **Problem Statement Title -** IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.
- **Theme -** MedTech / BioTech / HealthTech
- **PS Category -** Software
- **Team ID -** 152420
- **Team Name –** Team Code Monks

Code Monks

IP-SAKTI Sahayak: AI-Powered IPR & Regulatory Guidance for Ayurveda

SMART INDIA HACKATHON 2026

Problem:

This problem affects **multiple stakeholder**s across the Ayurvedic ecosystem:

- **Startups: Complex IPR and regulatory laws** delay or prevent safe **commercialization.**
- **Biopiracy Disputes: Scattered Traditional Knowledge** (TK) data enables **global patent theft.**
- **Local Communities: Unaware** of their rights on **ABS compliance and GI protection.**
- **Researchers & Students:** Face a **steep learning curve** navigating **overlapping Ayurvedic** legal frameworks.

Our Idea :

A **multilingual AI assistant** providing trusted, **source-cited** IPR and regulatory guidance for Ayurveda.

- **For Startups: Classifies formulations** to map applicable laws and **determine patentability.**
- **For Biopiracy Prevention**: **Cross-references TK databases** to flag and **block illegitimate commercialization.**
- **For Communities &Researchers:** Simplifies access to **IP rights, compliance requirements, and legal education.**

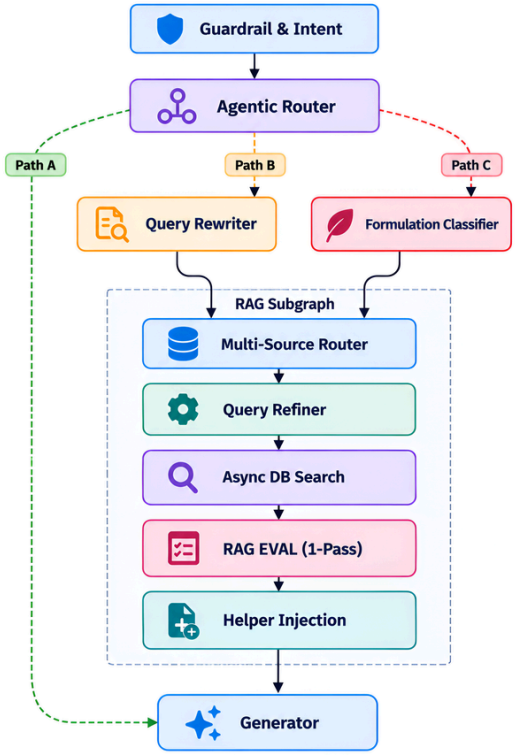
Proposed Solution : Agentic RAG System

1. The Gatekeepers (Safety & Routing)

- **Guardrail Assessor** instantly blocks out-of-scope requests to ensure **safe abstention** and **prevent hallucinations** (e.g., rejecting a legal advice, hypothetical lawsuit or a recipe).
- **Agentic Router analyzes user input** alongside **short-term memory** to dynamically direct traffic down the right path.

2. The 3-Way Execution Split

- **Path A** answers directly from **memory** for simple follow-ups (e.g., "Explain that simply") for fast responses with no DB search.
- **Path B** rewrites **vague queries** (e.g., resolving "How do I register it?") into specific legal searches, preserving classified state.
- **Path C** engages a **classifier** for new queries, asking clarifying questions to categorize the product (e.g. Classical vs. Proprietary medicine).



3. The RAG Subgraph (Scatter-Gather Retrieval)

- **Multi-Source Router** selects only the **relevant legal namespaces** (e.g., Patents Act, FSSAI) based on the classification to keep the search space clean.
- The **Query Refiner** translates user intent into **optimized search queries** tailored for each database. (Example: If both Patents and Biodiversity are selected, it creates two unique semantic queries tailored to how those specific laws are written).
- Next, **Async DB Search & CRAG Eval** fetch **legal chunks in parallel for speed** and rapidly verify the text actually contains the answer.
- Finally, a **Helper Injection** deterministically adds mandatory TKDL/ABS compliance warnings based on the product's classification.

4. Final Output Generation

- **The Synthesizer:** The main LLM heavy-lifter processes the validated context to **deliver a clean response, strictly formatted with clickable source citations** and an overall **system confidence score.**

TECHNICAL APPROACH

Technologies Used:

Agentic AI, Graph & Evaluation

- Text Extraction: 4LLM, Docling
- Agentic Workflow: LangChain, LangGraph
- Audit & Observability: LangSmith, LangStudio

Backend

- RAG Backend Pipeline: node.js, express.js
- Primary Database: NEON, PostgreSQL
- Multilingual Support: Bhashini API

RAG and LLM Layer

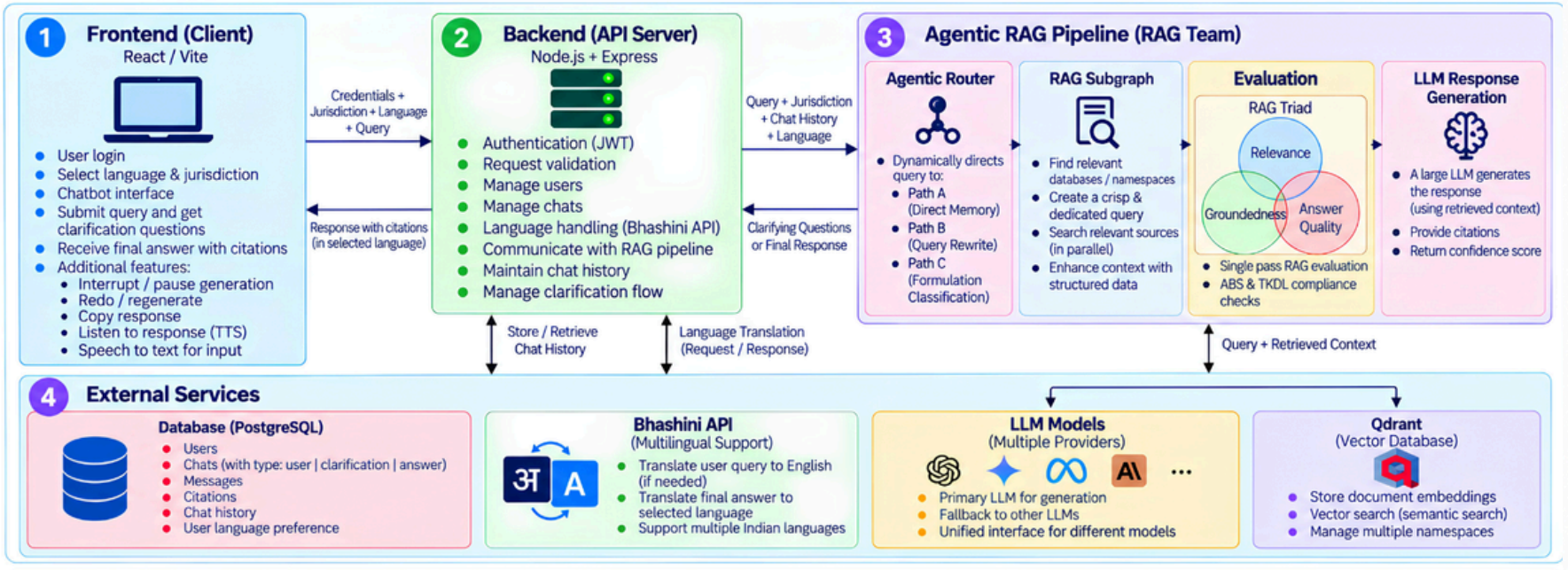
Fallback Models: groq, OpenAI, Google

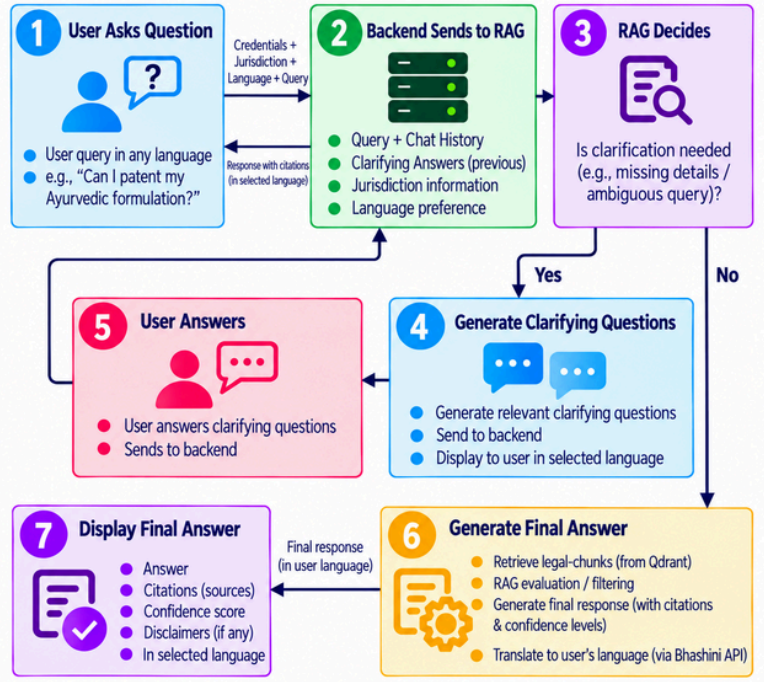
- Grounded Answers: RAG
- Query Embeddings: Google GenAI Embeddings
- Vector Database: Qdrant

Frontend

- PDF / Document Upload: File API + FormData
- Styling: Tailwind CSS
- Type Safety: TypeScript
- React Icons: Lucide

IP-SAKTI Sahayak System Architecture





FEASIBILITY AND VIABILITY

Potential Challenges & Risks

1. **Authoritative data sources** (e.g., TKDL) are often paywalled or **hard to access**, limiting corpus completeness.
2. Improving answer accuracy through deeper retrieval and evaluation **increases latency, risking user patience.**
3. Long conversations risk **context drift**, causing **token overload** and **misrouted queries.**
4. **Fast-changing IP law** (2024 Patent Rules, WIPO GRATK Treaty) can **outdate** a static corpus.
5. **Conflating** Indian and international law, or citing non-existent sources, **risks misleading answers**

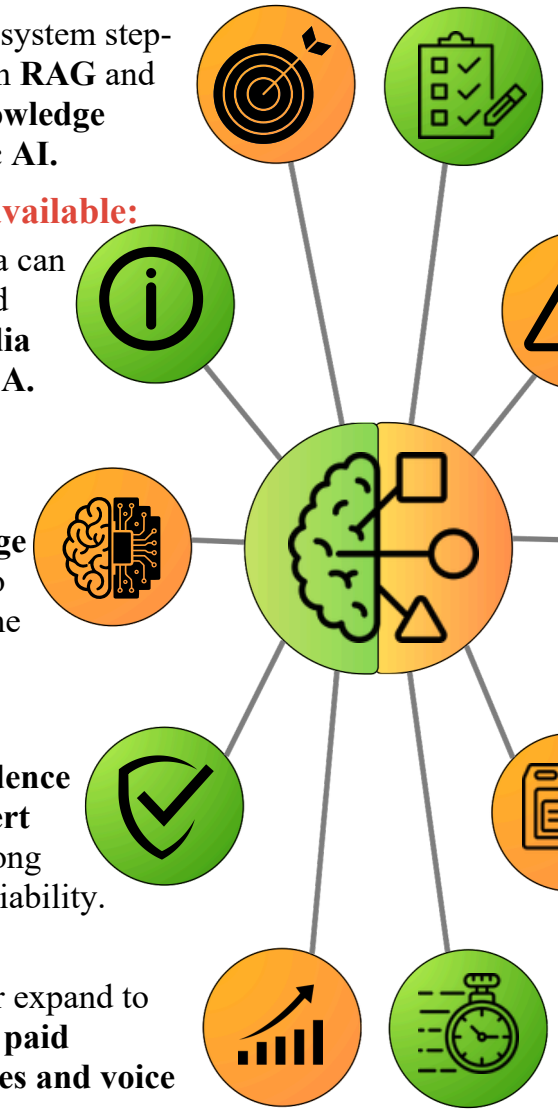
1. **Easy to build:**
We can develop the system step-by-step, starting with **RAG** and then adding the **Knowledge Graph and Agentic AI.**

2. **Reliable information available:**
Laws and Ayurveda data can be collected from trusted sources like **TKDL, India Code, IP India and NBA.**

3. **Practical technology:**
Bhashini, OCR, RAG, LangGraph and Knowledge Graph can work together to classify products and find the right legal information.

4. **Safe and testable:**
Source citations, confidence scores and human expert support help reduce wrong answers and improve reliability.

5. **Scalable solution**:
The system can later expand to more **legal sources, paid databases, languages and voice support.**



1. **Future-Proof Data:**
Version-tracked metadata keeps the corpus **current, absorbing new amendments and rules** without a full rebuild.

2. **Risk Mitigation:**
Guardrails against **hallucination, jailbreaks, and context drift** ensure reliability.

3. **Scalability:**
Easily expands to **new legal sources, languages,** and **voice support - —** architected to grow without disrupting the core pipeline.

4. **Sustainability:**
Free official sources (like India Code) keep the system **low-cost** and **self-sustaining.**

5. **Government Alignment:**
Directly supports **Ministry of AYUSH's mandate** to protect **traditional knowledge** and **enable safe commercialization.**

Strategies to overcome

1. MVP uses **free official sources** (nbaindia.org, ipindia.gov.in); **paid databases** can be **added later** via the user's own subscription, with permission.
2. A **fast router** handles **simple queries** instantly; **deeper verification** kicks in only for complex or uncertain ones.
3. **Short-term memory** plus **smart query rewriting** keeps follow-up questions accurate and on-topic.
4. **Legal data is** tagged by **version/year,** so the system always uses the current law — refreshed as amendments arrive.
5. **Legal data** is **tagged by country,** keeping Indian and international law **separate** — with every answer **fact-checked** before it's shown.

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IMPACT AND BENEFITS

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Impacts:

Ayurvedic Researchers & Innovators

Early patentability clarity, reducing wasted **R&D** and **legal effort** while **accelerating new formulation** discovery.

AYUSH Startups & MSMEs

Clear **FSSAI** and **advertising compliance** before **market launch,** avoiding **costly penalties** and **product recalls.**

Farmers & Tribal Cooperatives

Awareness of **legal benefit-sharing rights** over **traditional knowledge,** ensuring **fair compensation** and **community protection**

Government & Ministry of AYUSH

Scalable support to protect India's **traditional knowledge** with **evidence-backed oversight.**

Benefits:

Social

- Prevents **misuse** of **traditional knowledge.**
- Offers **multilingual legal guidance.**
- Empowers **farmers and tribal communities.**

Economic

- Cuts wasted **R&D** and **legal costs.**
- Prevents **penalties** and **product recalls.**
- Supports **safer commercialization.**

Legal & Governance

- Gives **source-cited legal guidance.**
- Covers **jurisdiction-specific rules** and **regulations.**
- Supports **ABS** and **TKDL compliance.**

Accessibility

- **Simplifies** complex **IPR** information.
- Explains **regulatory rules** in plain language.
- Gives **easy access** using **multilingual voice assistant.**

Innovation/uniqueness:

1. Future-Ready ETL Pipeline — **highly scalable architecture** built for **rapid ingestion** of new laws. acts and rules.

- ***Advanced Extraction:*** Utilizes **OCR and Markdown conversion** to perfectly extract text from *even camera-scanned legal PDFs.*
- ***Semantic Chunking***: Preserves **structural context** by keeping entire **Act sections** and **tables intact,** preventing broken data.
- ***Rich Metadata Tagging:*** Every chunk is tagged **with jurisdiction, source authority, act, chapter, and year**—*enabling precise, filtered retrieval.*
- ***Intelligent Filtering:*** Automatically **purges noise** (ToC, signatures) and retains only the **latest amendments**, smoothly overriding older rules.

2. ***End-to-End Observability*:**

- ***LangSmith Integration***: Features built-in audit capabilities, **end-to-end tracing**, prompt versioning, and automated **RAG Triad evaluations** to **catch and prevent hallucinations before they reach the user.**

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RESEARCH AND REFERENCES

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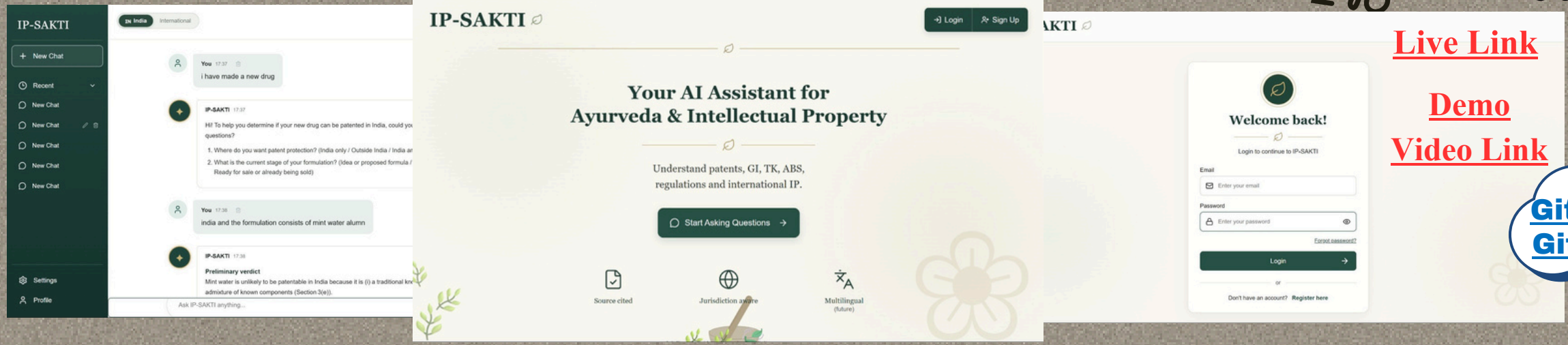
Research and Legal References

- **Ayurvedic Formulary of India (AFI)** Link
- **IP Laws and Geographical Indications** Link
- **AYUSH Drug Regulation** Link
- **Ayurveda-Aahar / FSSAI** Link
- **International Biodiversity** Link
- **International IP** Link
- **Traditional Knowledge** Link
- **Biodiversity & ABS** Link
- **Plant Variety Rights** Link

Future Ideas & Improvements:

1. ***Admin Dashboard — Data Management:*** **File-explorer UI** to visualize **namespaces** and **manage rules, acts,** and sections without touching code.
2. ***Admin Dashboard — Audit & Observability***: **In-app tracing** and **time-travel debugging** to replay any past query's **reasoning path.**
3. ***Static Helper Injection:*** **Deterministic, non-LLM ABS/TKDL** checks to **reduce load** on the **Generator node**.
4. ***Automated Answer Evaluation:*** **Post-generation checks** that automatically score **response accuracy.**
5. ***JEV Integration***: Enhances **classification accuracy** inside the **Formulation Classifier node** and **Agentic router node.**
6. ***Knowledge Graph Integration:*** Maps **relationships** between acts and sections for **deeper legal reasoning.**

Prototype



Live Link

Demo Video Link

GitHub GitHub